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Arrays for delivering tumor treating fields (TTFields) with sets of electrode elements having individually adjustable active areas

Abstract

Tumor treating fields (TTFields) can be delivered to a subject's body using electrode elements that are arranged in sets, wherein each set includes a respective first electrode element and a respective second electrode element disposed in thermal contact with each other. Individual first conductors provide an electrically conductive path between each of the first electrode elements and a respective pin of a connector. And a second conductor provides an electrically conductive path between all of the second electrode elements and another pin of the connector. Temperature sensors are disposed in thermal contact with each set of electrode elements. Because the electrode elements are arranged in sets, the current that flows through any given set can be reduced (with respect to its maximum value) by switching off the first electrode element within the given set, in order to prevent the area that corresponds to the given set from overheating.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This Application claims the benefit of U.S. Provisional Application 63/150,425, filed Feb. 17, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) TTFields therapy is a proven approach for treating tumors. FIG. 1 is a schematic representation of the prior art Optune® system for delivering TTFields. The TTFields are delivered to patients via four transducer arrays **21-24** that are placed on the patient's skin in close proximity to a tumor (e.g., as depicted in FIGS. 2A-2D for a person with glioblastoma). The transducer arrays **21-24** are arranged in two pairs, and each transducer array is connected via a multi-wire cable to an AC signal generator **20**. The AC signal generator (a) sends an AC current through one pair of arrays **21, 22** during a first period of time, which induces an electric field with a first direction through the tumor; then (b) sends an AC current through the other pair of arrays **23, 24** during a second period of time, which induces an electric field with a second direction through the tumor; then repeats steps (a) and (b) for the duration of the treatment.

(2) Each transducer array **21-24** is configured as a set of capacitively coupled electrode elements E

(e.g., a set of 9 electrode elements, each of which is about 2 cm in diameter) that are interconnected via a flex circuit. Each electrode element includes an electrically conductive substrate with a dielectric layer (more specifically, a layer of ceramic material with a high dielectric constant) disposed thereon. Each electrode element is sandwiched between a layer of an electrically conductive medical gel and an adhesive tape. When placing the arrays on the patient, the medical gel conforms to the contours of the patient's skin and ensures good electrical contact of the device with the body. The adhesive tape holds the entire array in place on the patient as the patient goes about their daily activities.

(3) The amplitude of the alternating current that is delivered via the transducer arrays is controlled so that skin temperature (as measured on the skin below the transducer arrays) does not exceed a safety threshold of 41° C. The temperature measurements on the patient's skin are obtained using thermistors T placed beneath some of the disks of the transducer arrays. In the existing Optune® system, each array includes 8 thermistors, with one thermistor positioned beneath a respective disk in the array. (Note that most arrays include more than 8 disks, in which case the temperature measurements are only performed beneath a sub-set of the disks within the array).

(4) The AC signal generator **20** obtains temperature measurements from all 32 thermistors (4 arrays×8 thermistors per array), and the controller in the AC signal generator uses the temperature measurements to control the current to be delivered via each pair of arrays in order to maintain temperatures below 41° C. on the patient's skin. The current itself is delivered to each array via an additional wire (i.e., one wire **28** for each of the arrays **21-24**) that runs from the AC signal generator **20** to each array. And an additional wire (not shown) for each of the arrays **21-24** is used as a common return for all 8 thermistors. Thus, each of the four cables that terminate on the arrays **21-24** in the existing Optune system has a total of 10 conductors.

SUMMARY OF THE INVENTION

(5) One aspect of the invention is directed to a first apparatus for applying an alternating electric field to a subject's body. The first apparatus comprises at least four sets of electrode elements, a connector, at least four first conductors, a second conductor, at least four temperature sensors, and a support configured to hold the sets of electrode elements against the subject's body. Each of the sets of electrode elements includes a respective first electrode element and a respective second electrode element disposed in thermal contact with the respective first electrode element. The connector has at least four first pins and a second pin. Each of the at least four first conductors provides an electrically conductive path between (a) a respective one of the first pins and (b) a respective one of the first electrode elements. The second conductor provides an electrically conductive path between the second pin and all of the second electrode elements; and each of the at least four temperature sensors is disposed in thermal contact with a respective one of the sets of electrode elements.

(6) In some embodiments of the first apparatus, within each of the sets of electrode elements, the area of the respective second electrode element is at least double the area of the respective first electrode element.

(7) In some embodiments of the first apparatus, the apparatus has at least nine sets of electrode elements, the connector has at least nine first pins, the apparatus has at least nine first conductors, and the apparatus has at least nine temperature sensors.

(8) In some embodiments of the first apparatus, each of the temperature sensors comprises a thermistor having a first terminal and a second terminal, the connector has a third pin, and each of the first conductors provides an electrically conductive path between (a) a respective one of the first pins, (b) a respective one of the first electrode elements, and (c) the first terminal of a respective thermistor. In these embodiments, the apparatus further comprises a third conductor that provides an electrically conductive path between the third pin and the second terminal of at least one of the thermistors. Optionally, in these embodiments, the second terminals of all the thermistors are wired together.

(9) In some embodiments of the first apparatus, each of the temperature sensors comprises a thermistor having a first terminal and a second terminal, the connector has a third pin, and each of the first conductors provides an electrically conductive path between (a) a respective one of the first pins, (b) a respective one of the first electrode elements, and (c) the first terminal of a respective thermistor. In these embodiments, the apparatus further comprises a third conductor that provides an electrically conductive path between the third pin and the second terminal of at least one of the thermistors. In these embodiments, the thermistors are wired in series, beginning with a first one of the thermistors and ending with a last one of the thermistors. The second terminal of each of the thermistors except for the last thermistor is wired to the first terminal of a respective subsequent thermistor, and the third conductor provides an electrically conductive path between the third pin of the connector and the second terminal of the last thermistor.

(10) In some embodiments of the first apparatus, each of the temperature sensors comprises a region of a pyroelectric material.

(11) In some embodiments of the first apparatus, each of the first electrode elements comprises a conductive plate with a dielectric layer disposed thereon, each of the second electrode elements comprises a conductive plate with a dielectric layer disposed thereon, and the support is configured to hold the first electrode elements and the second electrode elements against the subject's body so that the dielectric layers of the first electrode elements and the dielectric layers of the second electrode elements face the subject's body.

(12) Another aspect of the invention is directed to a second apparatus for applying an alternating electric field to a subject's body using at least four sets of electrode elements, wherein each of the sets of electrode elements includes a respective first electrode element and a respective second electrode element disposed in thermal contact with the respective first electrode element, and wherein each of the sets of electrode elements is disposed in thermal contact with a respective temperature sensor. The second apparatus comprises an AC signal generator that generates an AC output signal. The second apparatus also comprises a connector that includes at least four first pins and a second pin, wherein each of the first pins corresponds to a respective one of the first electrode elements, and wherein the AC output signal is applied to the second pin. The second apparatus also comprises at least four first switches, wherein each of the first switches is configured to selectively either apply or not apply the AC output signal to a respective one of the first pins depending on a state of at least one control signal. The second apparatus also comprises an amplifier configured to accept an input from each of the temperature sensors and generate a corresponding output. And the second apparatus also comprises a controller configured to, based on the output of the amplifier, set the at least one control signal to a state that determines whether the AC output signal is applied or not applied to each of the first pins.

(13) In some embodiments of the second apparatus, the controller is configured to (a) determine, based on the output of the amplifier, when at least one of the first electrode elements is hotter than other first electrode elements and (b) set the at least one control signal to a state that controls the first switches so that the AC signal is not applied to at least one respective first pin.

(14) In some embodiments of the second apparatus, the controller is configured to (a) determine, based on the output of the amplifier, when at least one of the first electrode elements is hotter than a threshold level and (b) set the at least one control signal to a state that controls the first switches so that the AC signal is not applied to at least one respective first pin.

(15) In some embodiments of the second apparatus, the inputs from the temperature sensors arrive via the same first pins that correspond to the first electrode elements.

(16) Another aspect of the invention is directed to a first method of applying an alternating electric field to a subject's body. The first method comprises positioning at least four sets of electrode elements on or in the subject's body, wherein each of the sets of electrode elements has an active area that is adjustable. The first method also comprises energizing each of the sets of electrode elements using its entire active area; measuring a respective temperature of each of the sets of

electrode elements; and reducing the active area of at least one of the sets of electrode elements based on a corresponding one of the temperature measurements.

(17) In some instances of the first method, the active area of a given set of electrode elements is reduced when the given set of electrode elements is hotter than other sets of electrode elements. In some instances of the first method, the active area of a given set of electrode elements is reduced when the given set of electrode elements is hotter than a threshold level.

(18) Another aspect of the invention is directed to a second method of applying an alternating electric field to a subject's body. The second method comprises positioning at least four sets of electrode elements on or in the subject's body, wherein each of the sets of electrode elements has an active area that is adjustable. The second method also comprises energizing each of the sets of electrode elements using its entire active area; measuring a respective temperature of each of the sets of electrode elements; and reducing the active area of at least one of the sets of electrode elements based on a corresponding one of the temperature measurements. The positioning comprises positioning at least four first electrode elements on or in the subject's body and positioning at least four second electrode elements on or in the subject's body. Each of the first electrode elements is wired so that it can be energized independently of the other first electrode elements. Each of the second electrode elements is positioned adjacent to and in thermal contact with a respective one of the first electrode elements. The second electrode elements are wired together so that all of the second electrode elements must be either collectively energized or collectively not energized. The energizing comprises energizing the first electrode elements and all of the second electrode elements. The reducing of the active area comprises deenergizing selected ones of the first electrode elements based on a respective temperature measurement.

(19) In some instances of the second method, the deenergizing of a given first electrode element occurs when the given first electrode element is hotter than other first electrode elements.

(20) In some instances of the second method, the deenergizing of a given first electrode element occurs when the given first electrode element is hotter than a threshold level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic representation of the prior art Optune® system for delivering TTFields.

(2) FIGS. 2A-2D depict the positioning of transducer arrays on a person's head for treating a brain tumor.

(3) FIG. 3 is a schematic representation of a transducer array that is used for applying TTFields to a subject's body.

(4) FIG. 4A-4C depict three respective approaches for positioning temperature sensors in thermal contact with the electrode elements in a transducer array.

(5) FIG. 5 depicts a transducer array that provides individual control over the current that passes through nine different areas of the transducer array.

(6) FIG. 6 is a block diagram of a system that uses four copies of the FIG. 5 transducer array to apply TTFields to a subject.

(7) FIG. 7 depicts another transducer array that provides individual control over the current that passes through nine different areas of the transducer array.

(8) FIG. 8 is a block diagram of a system that uses four copies of the FIG. 7 transducer array to apply TTFields to a subject.

(9) FIG. 9 is a schematic diagram of a circuit that is suitable for implementing each of the switches in banks 1L and 1R in the FIG. 6 and FIG. 8 embodiments.

(10) Various embodiments are described in detail below with reference to the accompanying drawings, wherein like reference numerals represent like elements.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(11) Although the FIG. 1 approach described above is very effective for delivering TTFields to a tumor, the effectiveness of the treatment will drop if good electrical contact is not maintained between each of the elements in the four transducer arrays **21-24** and the person's body. This can happen, for example, if the hydrogel beneath one or more elements of the transducer arrays dries out over time, or due to hair growth beneath one or more of the elements.

(12) Assume, for example, that there are 9 electrode elements **E** in each of the transducer arrays **21-24**, that the hydrogel beneath a single electrode element **E** on the front transducer array **21** has dried out; and that enough hydrogel is present beneath (a) all the other electrode elements **E** of that transducer array **21**, and (b) all the electrode elements **E** of the other transducer arrays **22-24**. In this situation, the resistance between the single electrode element **E** and the person's body will be higher than the resistance between any of the other electrode elements and the person's body. And this increase in resistance will cause the temperature of the single electrode element **E** to rise more than the other electrode elements.

(13) In this situation, because all of the electrode elements **E** in each of the transducer arrays **21-24** are wired in parallel, the AC signal generator **20** must limit the current that is applied to the entire front/back pair of transducer arrays **21, 22** in order to keep the temperature of the single electrode element **E** on the front array **21** below 41°, even though the temperature at all the remaining electrode elements **E** on the front and back transducer arrays **21, 22** may be well below 41° C. And this decrease in current causes a corresponding decrease in the strength of the electric field at the tumor, which can reduce the efficacy of the treatment.

(14) One possible approach for dealing with this situation is to wire a separate conductor to each of the 9 electrode elements (as opposed to the prior art approach of wiring all of the electrode elements together in parallel). If this approach is implemented, it becomes possible to switch off the AC signal to whichever electrode element is overheating without switching off the AC signal to the other electrode elements on the same array. This approach is referred to herein as the “individually addressable electrode approach.”

(15) But switching an electrode element completely off using the individually addressable electrode approach can cause the current that passes through the remaining electrode elements to increase, which would raise their temperature. In addition, switching an electrode element completely off could have a negative impact on the distribution of the electric field within the subject's body. Moreover, the inventor has determined that in the vast majority of situations, a current reduction of less than 20% will prevent any given electrode element from overheating. As a result, switching an electrode element completely off using the individually addressable electrode approach could be considered to be overkill. The embodiments described below obviate or minimize the problems identified in this paragraph by replacing each of the prior art electrode elements with a set of electrode elements.

(16) FIG. 3 is a schematic representation of a transducer array **50** that includes nine sets **52/53** of electrode elements that are used for applying TTFields to a subject's body. Each set includes a first electrode element **52** and a second electrode element **53** that are disposed in thermal contact with each other. The overall array **50** includes at least four sets **52/53** of first and second electrode elements (e.g., nine sets **52/53** in the embodiment illustrated in FIG. 3, or another number between 4 and 50). A separate first conductor is wired to each of the first electrode elements **52**, which makes it possible to switch the AC signal to any given one of the first electrode elements **52** on or off independently. But all of the second electrode elements **53** are wired in parallel to a second conductor, which means that whenever an AC signal is applied to the second conductor, the AC signal will reach all of the second electrode elements **53**.

(17) In some preferred embodiments, the second electrode element **53** within any given set is at least double the area of the respective first electrode element **51**. Assume, for purposes of discussion, that within any given set of electrode elements **52/53**, 70% of the total area is occupied

by the second electrode element **53**, and 30% of the total area is occupied by the first electrode element **52**. When delivering TTFields, the current that passes through any given set **52/53** of electrode elements is related to the active area of that set. As a result, when a given AC voltage is applied to both the first electrode element **52** and the second electrode element **53**, a full measure of current (i.e., 100%) will pass through the set **52/53**. But when the same AC voltage is applied to only the second electrode element **53** and not to the first electrode element **52**, the current that passes through the entire set **52/53** will drop from 100% to a lower level (e.g., 80%).

(18) Assume that an AC voltage is applied to the second electrode element **53** during an interval of time (e.g., a 1 second interval of time) via the second conductor. Further assume that during this same interval of time, the same AC voltage is applied to the first electrode element **52** in the set of electrode elements **52/53** labeled X (via a corresponding first conductor), but the AC voltage is not applied to the first electrode element **52** in the set of electrode elements **52/53** labeled Z. In this situation, because current is related to the active area, a full measure of current (i.e., 100%) will pass through the set **52/53** labeled X, but a lower current will pass through the second set **52/53** labeled Z.

(19) The first electrode element **52** and a second electrode element **53** within any given set **52/53** are shaped and positioned so as to be in thermal contact with each other (i.e., they are shaped and positioned so that heating up the first electrode element **52** will cause the second electrode element **53** to heat up, and vice versa). Note that the thermal contact between the first electrode element **52** and the second electrode element **53** may be indirect thermal contact, with intervening components disposed between the first electrode element **52** and the second electrode element **53**. One preferred approach for achieving thermal contact between the first and second electrode elements **52, 53** within any given set is to shape those electrode elements **52, 53** as interleaved spirals (not shown) or as interleaved squared-off spirals (as depicted in FIG. 3). In alternative embodiments, different interleaved patterns may be used (e.g., interleaved stripes or interleaved comb-shaped patterns). Interleaving the first and second electrode elements **52, 53** in this manner will improve the thermal contact between the first and second electrode elements **52, 53**, which will minimize the variation in temperature between those elements.

(20) A temperature sensor is disposed in thermal contact with each set **52/53** of electrode elements. (Here again, the thermal contact may be indirect.) Preferably, the number of temperature sensors matches the number of sets of electrode elements. For example, when four sets **52/53** of electrode elements are used, there will be four temperature sensors. In some embodiments, thermistors are used as the temperature sensors.

(21) FIG. 4A depicts a first approach for positioning the temperature sensor (e.g., a thermistor **54**) in thermal contact with the first and second electrode elements **52, 53**. In this approach, a thermistor **54** is positioned in an open space between the first and second electrode elements **52, 53** within each set **52/53**, so that the thermistor **54** is disposed in thermal contact with both the first and second electrode elements **52, 53**. FIG. 4B depicts a second approach for positioning the temperature sensor. In this approach, the first and second electrode elements **52, 53** occupy almost the entire area of the set **52/53**, and a thermistor **54** is positioned on a backside of the set **52/53**, so that the thermistor **54** is disposed in thermal contact with both the first and second electrode elements **52, 53**. This approach is particularly well suited in cases where the first and second electrode elements **52, 53** are implemented using respective traces of a flex circuit. Two approaches for using a set of thermistors as the temperature sensors are described below in connection with FIGS. 5-8.

(22) FIG. 4C depicts a third approach for positioning the temperature sensor. In this approach, the first and second electrode elements **52, 53** occupy almost the entire area of the set **52/53**, and temperature-sensing is implemented by positioning regions of a pyroelectric material (not shown) behind and in thermal contact with both the first and second electrode elements **52, 53**. This approach is also particularly well suited in cases where the first and second electrode elements **52,**

53 are implemented using respective traces of a flex circuit. A description of how to use regions of pyroelectric material to sense temperature is provided below.

(23) The embodiments described herein advantageously provide the ability to reduce the current that flows through a given area of a transducer array 50, without completely shutting off the current that flows through the given area.

(24) One approach for controlling the current that passes through each area of a transducer array is to (1) start off with the prior art configuration depicted in FIG. 1; (2) reconfigure each of the electrode elements E to resemble the shape of the second electrode elements 53 depicted in FIG. 3; (3) add nine additional electrode elements that are shaped like the first electrode elements 52 depicted in FIG. 3 and interleaved with the original electrode elements E; and (4) add an additional conductor that runs to each of the new electrode elements so that the new electrode elements can be energized individually. While this approach is workable, it requires almost double the number of conductors in each of the cables that runs to the transducer arrays. For example, in transducer arrays that have 9 controllable areas, a total of 20 wires would be needed in each cable (i.e., 1 to provide common access to all of the original electrode elements, 9 to provide individual access to each of the 9 new electrode elements, an additional 9 for the signals from the thermistors, plus one additional wire to serve as a common return for all 9 thermistors). And this significant increase in the number of wires in each cable tends to make the cables less flexible and more cumbersome, which can make the system harder to use, and reduce patient compliance.

(25) The embodiments described below advantageously provide the ability to control the current that is routed through individual areas of the transducer array 50, without unduly increasing the number of conductors in the cables that terminate on the transducer arrays. These embodiments can be used to implement systems that distribute the functions of outputting current (in order to generate TTFields) and obtaining temperature readings into mutually exclusive time slots or phases. In these systems, because temperature readings are not obtained at the exact same instant of time while the transducer arrays are outputting current, the same set of conductors can be used to output current and to input temperature readings. This advantageously reduces the total number of conductors that must be included in each cable.

(26) One suitable approach for distributing the functions of outputting current and obtaining temperature readings into mutually exclusive time slots or phases is to (i) output current for an interval of time (e.g., 1 s) and then turn the current off, then (ii) spend a short period of time (e.g., 10 ms) obtaining temperature measurements, and then repeating those two steps (i) and (ii) in an alternating sequence repeatedly (e.g., for 12-18 hours per day). The operation of the system during each of those phases (i.e., the current-outputting phase and the temperature-reading phase) is described below for a variety of embodiments.

(27) FIG. 5 is a schematic representation of a first embodiment of a transducer array 50 that provides individual control over the current that passes through nine different areas of the transducer array 50 during the current-outputting phase. As will be described below in connection with FIG. 6, four copies of the transducer array 50 are preferably used to administer TTFields treatment to a person's head (or other body part).

(28) Each transducer array 50 includes at least four sets 52/53 of electrode elements. Each set 52/53 includes a respective first electrode element 52 and a respective second electrode element 53 disposed in thermal contact with each other. The first electrode elements 52 are labeled E1-E9 and the second electrode elements 53 are labeled A1-A9 for ease of reference in the FIG. 5 embodiment. The shape and positioning of the first and second electrode elements 52, 53 with respect to each other is as described above in connection with FIGS. 3-4. (Note that the shape and positioning is not depicted in FIG. 5 to make it easier to see the electrical interconnections between the various components.) Each set 52/53 of electrode elements is positioned at a different area of the transducer array 50, and the first and second electrode elements 52, 53 within any given set are positioned in thermal contact with each other.

(29) Each of the first and second electrode elements **52**, **53** has an electrically conductive substrate with a dielectric layer disposed thereon. The electrically conductive substrate may be implemented using a thin layer of metal. The dielectric layer may be implemented using a ceramic material or a layer of a polymer with a high dielectric constant (e.g., at least 20).

(30) In the embodiment depicted in FIG. 5, all of the first and second electrode elements **52**, **53** are held in place by a support structure **59**. The support structure is configured to hold the electrode elements against the subject's body so that the dielectric layer of the first and second electrode elements **52**, **53** faces the subject's body and can be positioned in contact with the subject's body. Optionally, this support structure may comprise a flexible backing **59** (e.g., a layer of foam material). Preferably, a layer of hydrogel is disposed between the dielectric layer of the first and second electrode elements **52**, **53** and the subject's body when the transducer array **50** is placed against the subject's body. Construction of the support structure **59** may be implemented using any of a variety of conventional approaches that will be apparent to persons skilled in the relevant arts, including but not limited to self-adhesive fabric, foam, or plastic sheeting.

(31) Each transducer array **50** also has a connector **57** that is used to send electrical signals into and out of the transducer array **50**. The connector **57** has at least four first pins and a second pin. In the illustrated embodiment, the number of first pins is the same as the number of first electrode elements **52**, and each of the first pins corresponds to a respective one of those first electrode elements **52**. And in the illustrated embodiment, there is only a single second pin, labeled A. Note that as used herein, the term “pin” can refer to either a male or female pin of the connector **57**.

(32) Each of the first electrode elements **52** (labeled E1-E9) is wired via a respective individual first conductor to a respective first pin of the connector **57**. More specifically, each of the first conductors provides an electrically conductive path between (a) a respective one of the first pins in the connector **57** and (b) the conductive substrate of a respective one of the first electrode elements **52** (E1-E9). These first conductors are numbered 1-9 just above the “wire routing” block **55** (which funnels the individual conductors together into a single cable **56**). In some preferred embodiments, the electrical connection to each of the first electrode elements **52** comprises one or more traces on a flex circuit and/or one or more conductive wires.

(33) Because the connector **57** has an individual first pin that corresponds to each of the individual first electrode elements **52**, and because an electrically conductive path exists between each of the first pins and a respective one of the first electrode elements **52**, the system that mates with the connector **57** can selectively energize or not energize each of the first electrode elements **52** individually by either applying or not applying an AC signal to the respective first pin on the connector **57**. In contrast, all of the second electrode elements **53** (labeled A1-A9) are connected via an electrically conductive path (e.g., wired together in series or parallel) to the node labeled “A” which eventually terminates on a second pin of the connector **57**. As a result, the system that mates with the connector **57** must selectively energize or not energize all of the second electrode elements **53** together by either applying or not applying an AC signal to the second pin on the connector **57**.

(34) Therefore, when the system that mates with the connector **57** is applying an AC signal to the second pin on the connector **57**, any given individual area of the transducer array **50** will either (a) pass a full level of current (i.e., when the respective first electrode element **52** is energized) or (b) pass a lower level of current (i.e., when the respective first electrode element **52** is not energized). This advantageously provides the ability to reduce the current that flows through a given area of a transducer array **50** without completely shutting off the current that flows through that area.

(35) We shall now discuss operation of the FIG. 5 embodiment during the temperature-reading phase. Each transducer array **50** also includes at least four temperature sensors, each of which is disposed in thermal contact with a respective one of the sets **52/53** of electrode elements. In the embodiment illustrated in FIG. 5, the temperature sensors are implemented using thermistors **54**, with one thermistor positioned with respect to each set **52/53** so that the thermistor **54** can sense the

temperature of the first and second electrode elements **52**, **53** within that set. This may be accomplished, for example, using any of the approaches described above in connection with FIGS. **4A-4B**. Each of the thermistors **54** has a first terminal (i.e., the lower terminal of the thermistor in FIG. **5**) and a second terminal (i.e., the upper terminal of the thermistor in FIG. **5**).

(36) In this embodiment, each of the first conductors provides an electrically conductive path between (a) a respective one of the first pins in the connector **57**, (b) the conductive substrate of a respective one of the first electrode elements **52** (**E1-E9**), and (c) the first terminal of the corresponding thermistor **54**.

(37) In the FIG. **5** thermistor-based embodiment, each transducer array **50** has a third conductor that provides an electrically conductive path between a third pin of the connector **57** (labeled C in FIG. **5**) and the second terminal (i.e., the upper terminal in FIG. **5**) of at least one of the thermistors **54**. In the embodiment depicted in FIG. **5**, the second terminal of all of the thermistors are wired together. In this embodiment, the third conductor provides an electrically conductive path between the third pin of the connector **57** and the second terminal of all of the thermistors **54**. The third conductor may optionally be implemented using a plurality of segments of wire and/or a plurality of traces on a flex circuit.

(38) Because the connector **57** has an individual first pin that corresponds to the first terminal of each of the thermistors **54** and because an electrically conductive path exists between each of the first pins and a respective one of the thermistors **54**, the system that mates with the connector **57** has access to the first terminal of each of the thermistors **54**. In addition, because the second terminal of all the thermistors **54** are all wired together and connected to the third pin (labeled C), the system that mates with the connector **57** also has access to the second terminal of each of the thermistors **54**. As a result, the system that mates with the connector **57** can measure the resistance of any of the thermistors **54** during the temperature-reading phase. This may be accomplished, for example, by routing a known current through each thermistor **54** and measuring the voltage that appears across each thermistor.

(39) Notably, because any given first pin on the connector **57** corresponds to a respective one of the individual first electrode elements **52** (during the current-outputting phase) and also corresponds to a respective one of the individual thermistors **54** (during the temperature-reading phase), each of the first pins on the connector **57** serves two functions. This reduces the number of wires that must be included in each of the cables **56**, which in turn advantageously makes the cables more flexible and less cumbersome.

(40) FIG. **6** is a block diagram of a system that uses four copies of the transducer array **50** (described above in connection with FIG. **5**) to apply TTFields to a subject. In FIG. **6**, these four copies are labeled **50A**, **50P**, **50L**, and **50R**, where A, P, L, and R stand for anterior, posterior, left, and right, respectively. The lower portion of FIG. **6** depicts an AC voltage generator **35** and a “CAD box” **30** as separate blocks, the latter of which includes a temperature measurement block **32**, a controller **34**, and banks of switches **1L**, **2L**, **3L**, **1R**, **2R**, and **3R**. In some embodiments, the components in those two blocks **35**, **30** may be physically divided into two separate housings. But in alternative embodiments, the components in those two blocks **35**, **30** are combined into a single housing.

(41) For clarity, only the left and right channels are depicted in FIG. **6**. But the remaining channels (i.e., the anterior and posterior channels) operate in the same way as the left and right channels, respectively. Additionally, each of the transducer arrays **50** in FIG. **6** is depicted with only 4 first electrode elements **52** and four thermistors **54** for clarity. But it is expected that practical systems will have a larger number (e.g., between 9 and 30) of first electrode elements and thermistors, and will also have a larger number of certain other components (e.g., switches, conductors, etc.), depending on the number of first electrode elements **52** that are actually used in each of the transducer arrays **50**.

(42) The FIG. **6** system can measure the temperatures of the thermistors **54** in the left channel **50L**

during the temperature-reading phase by controlling the electronically-controlled switches in bank 2L (which may be implemented using bidirectional analog switches) to select each of the thermistors in turn. For example, switch C and switch 1 should be closed to select thermistor T1; switch C and switch 2 should be closed to select thermistor T2; etc. After any given one of the thermistors T1-T4 within the transducer array 50L has been selected, the temperature measurement block (TMB) 32 can determine the temperature of the thermistor by measuring the resistance of the thermistor. This may be accomplished, for example, by using a current source that generates a known current (e.g., 150 μ A) positioned within the TMB 32, so that the known current will be routed into whichever thermistor is selected by the bank of switches 2L at any given instant. The known current will cause a voltage to appear across the selected thermistor (T1-T4), and the temperature of the selected thermistor can be determined by measuring this voltage. A controller 34 runs a program that selects each of the thermistors T1-T4 in turn and measures the voltage that appears across each of the thermistors (which is indicative of the temperature at the selected thermistor) in turn. An example of suitable hardware and procedures that may be used to obtain temperature readings from each of the thermistors is described in US 2018/0050200, which is incorporated herein by reference in its entirety.

(43) Measuring the temperature of the thermistors 54 in the right channel 50R is accomplished using the same approach described above in connection with the left channel 50L, except that the bank of switches 2R is used instead of bank 2L. Corresponding banks of switches (not shown) are also provided for the other channels 50A, 50P, and a similar approach is used in those channels as well.

(44) Assume that within a given interval of time (e.g., 1 second) during the current-outputting phase, the AC voltage generator 35 is applying an AC signal between the L and R terminals. This AC signal is applied to the A terminals of the left and right transducer arrays 50L, 50R, which means that the output of the AC voltage generator is applied to all of the second electrode elements 53 (A1-A4). Based on the temperature readings obtained from the thermistors 54 (T1-T4) during the temperature-reading phase, the controller 34 controls the switches in bank 1L to either turn on or turn off the current (which originates in the AC voltage generator 35) to each of the corresponding first electrode elements 52 (E1-E4) during the next current-outputting phase. For example, to leave the full current on for all four of the sets 52/53, all four of the switches in bank 1L should be closed. To reduce the current that passes through the E1/A1 set 52/53 of electrode elements (see FIG. 5), switch 1 in bank 1L should be opened; and to reduce the current that passes through the E2/A2 set 52/53 of electrode elements, switch 2 in bank 1L should be opened; etc.

(45) Controlling the current that is routed through individual first electrode elements during the current-outputting phase can therefore be used to reduce the current that passes through a given area when that area begins to get hot. This can advantageously prevent overheating without completely turning off the current that passes through a given area.

(46) In some embodiments, the controller 34 may be programmed to keep the temperature at all of the areas below a safety threshold (e.g., below 41° C.) as follows: Begin by closing all of the switches 1-4 in bank 1L, so that a full measure of current (i.e., 100%) passes through each set 52/53 of electrode elements during the current-outputting phase. Then, during the temperature-reading phase, based on signals arriving via the TMB 32, the controller 34 determines whether the temperature at each of the areas exceeds an upper threshold (e.g., 40° C.) that is below the safety threshold. When the controller 34 detects this condition, the controller 34 reduces the current that passes through the warmer areas by switching off the signal to the first electrode element 52 within those areas during the next current-outputting phase. Notably, this procedure only lowers the current that passes through certain areas of the transducer array 50, and does not lower the current that passes through the remaining areas on that transducer array 50.

(47) Optionally, the decision by the controller 34 to turn off a given first electrode element 52 may be based on the speed at which the corresponding thermistor 54 heats up (as measured via the

temperature sensors **54** and the TMB **32** during two or more temperature-reading phases that are spaced in time). More specifically, if the controller **34** recognizes that a given thermistor **54** is heating up faster than expected, the controller **34** can proactively open the switch that feeds the corresponding first electrode element **52** during subsequent current-outputting phases.

(48) Optionally, the controller **34** can control the current that passes through any given area on the transducer array **50** based on real-time temperature measurements. For example, if the temperature at a given area reaches 40° C., the controller **34** can open the switch that feeds the corresponding first electrode element **52**, which will reduce the current that passes through the corresponding area during the current-outputting phase. The controller **34** then waits until the temperature measured using the temperature sensors **54** drops below a second temperature threshold (e.g., below 38° C.). Once the temperature drops below this second temperature threshold, the controller **34** can close the switch that feeds the corresponding first electrode element **52**, which will restore the current that passes through the corresponding area during the current-outputting phase to its original value.

(49) Individually switching the current to each of the first electrode elements **52** in the right channel **50R** is accomplished using the same approach described above in connection with the left channel **50L**, except that the bank of switches **1R** is used instead of bank **1L**. Corresponding banks of switches (not shown) are also provided for the other channels **50A**, **50P**, and a similar approach is used in those channels as well.

(50) In some preferred embodiments, the AC signal generator **35** (a) sends an AC current through the anterior/posterior arrays **50A/50P** during a first period of time (e.g., 1 sec.), which induces an electric field with a first direction through the tumor in the subject's body; then (b) sends an AC current through the left/right arrays **50L/50R** during a second period of time (e.g., 1 sec.), which induces an electric field with a second direction through the tumor; then repeats steps (a) and (b) for the duration of the treatment. In these embodiments, the controller **34** may decide whether to switch each of the first electrode elements **52** on or off just prior to each 1 sec. interval of time.

(51) Optionally, an additional bank of switches **3L** may be provided. Each of the switches in this bank is wired in parallel with a corresponding one of the thermistors **T1-T4**, so that when a given one of the switches **1-4** is closed, a respective one of the thermistors **T1-T4** will be shorted out.

(52) The reason for including the additional bank of switches **3L** is that when (a) current from the AC voltage generator **35** is flowing through the first electrode elements **52** of the left channel **50L**, the subject's body, and the first electrode elements **52** of the right channel **50** during the current-outputting phase, and (b) the power to any of the first electrode elements **52** of the left channel **50L** is switched off by a corresponding one of the switches in bank **1L**, current can sneak through the thermistors **54** in the left channel **50L**. Assume, for example, that only switch #2 in bank **1L** is switched off (i.e., open). Because switches #1, 3, 4 are switched on (i.e., closed), the AC voltage generator **35** will impose a voltage on electrode elements **E1**, **E3**, **E4**. Thermistors **T1** and **T2** provide a path for current to flow from **E1** to **E2**; thermistors **T3** and **T2** provide a path for current to flow from **E3** to **E2**; and thermistors **T4** and **T2** provide a path for current to flow from **E4** to **E2**. This is equivalent to the parallel combination of **E1**, **E3**, and **E4** wired in series with **E2**. Because the number of thermistors in this parallel combination increases linearly with the number of first electrode elements **52**, the current in the single thermistor **E2** (which is wired in series with the parallel combination) can become significant. Including the optional additional bank of switches **3L** provides the system with the ability to prevent power dissipation in that single thermistor **E2** by closing the corresponding switch #2 in bank **3L**.

(53) To accomplish this (in those embodiments that include the additional bank of switches **3L**), the controller **34** may be programmed so that any time a given one of the switches in bank **1L** is opened, the corresponding switch in bank **3L** is closed. This will prevent the thermistor **54** associated with the switched-off first electrode element **52** from dissipating too much power, as described in the previous paragraph.

(54) In those embodiments that include the additional bank of switches **3L**, individual bypassing of

each of the thermistors **54** in the right channel **50R** is accomplished using the same approach described above in connection with the left channel **50L**, except that the bank of switches **3R** is used instead of bank **3L**. Corresponding banks of switches (not shown) are also provided for the other channels **50A**, **50P**, and a similar approach is used in those channels as well.

(55) FIG. 7 depicts a second embodiment of a transducer array **150** that provides individual control over the current that passes through nine different areas of the transducer array **150** during the current-outputting phase. As will be described below in connection with FIG. 8, four copies of the transducer array **150** are preferably used to administer TTFields treatment to a person's head (or other body part).

(56) Each transducer array **150** includes at least four sets **152/153** of electrode elements. Each set **152/153** includes a respective first electrode element **152** and a respective second electrode element **153** disposed in thermal contact with each other. The first electrode elements **152** are labeled E1-E9 and the second electrode elements **153** are labeled A1-A9 for ease of reference. The first and second electrode elements **152**, **153** are similar, respectively, to the first and second electrode elements **52**, **53** of the FIGS. 5-6 embodiment described above. Each set **152/153** of electrode elements is positioned at a different area of the transducer array **150**, and the first and second electrode elements **152**, **153** within any given set are positioned in thermal contact with each other.

(57) The first and second electrode elements **152**, **153** are held in place by a support structure **159**, which is similar to the support structure **59** in the FIG. 5 embodiment.

(58) Each transducer array **150** also has a connector **157** that is used to send electrical signals into and out of the transducer array **150**. The connector **157** has at least four first pins and a second pin. In the illustrated embodiment, the number of first pins is the same as the number of first electrode elements **152**, and each of the first pins corresponds to a respective one of those first electrode elements **152**. And in the illustrated embodiment, there is only a single second pin, labeled A. Note that as used herein, the term “pin” can refer to either a male or female pin of the connector **157**.

(59) Each transducer array **150** also has at least four first conductors, and the number of these first conductors will depend on the number of first electrode elements **152**. For example, in the embodiment depicted in FIG. 7, which contains 9 first electrode elements **152**, there are 9 first conductors. Each of these first conductors provides an electrically conductive path between (a) a respective one of the first pins in the connector **157** and (b) the conductive substrate of a respective one of the first electrode elements **152** (E1-E9). These first conductors are labeled 1-9 just above the “wire routing” block **155** (which funnels the individual conductors together into a single cable **156**). As in the FIG. 5 embodiment, each of these first conductors may optionally be implemented using a plurality of segments of wire and/or a plurality of traces on a flex circuit.

(60) As in the FIG. 5-6 embodiment, the system that mates with the connector **157** can selectively energize or not energize each of the first electrode elements **152** individually by either applying or not applying an AC signal to the respective first pin on the connector **157** during the current-outputting phase. But because all of the second electrode elements **153** are wired together, it must selectively energize or not energize all of the second electrode elements **153** together by either applying or not applying an AC signal to the second pin on the connector **157**. Therefore, any given individual area of the transducer array **150** will either (a) pass a full level of current (i.e., when the respective first electrode element **152** is energized) or (b) pass a lower level of current (i.e., when the respective first electrode element **152** is not energized).

(61) We shall now discuss operation of the FIG. 7 embodiment during the temperature-reading phase. Each transducer array **150** also includes at least four temperature sensors, each of which is disposed in thermal contact with a respective one of the sets **152/153** of electrode elements. As in the FIG. 5-6 embodiment, the temperature sensors may be implemented using thermistors **154**. Each of the thermistors **154** has a first terminal (i.e., the lower terminal of the thermistor in FIG. 7) and a second terminal (i.e., the upper terminal of the thermistor in FIG. 7). In these embodiments, each of the first conductors provides an electrically conductive path between (a) a respective one of

the first pins in the connector **157**, (b) the conductive substrate of a respective one of the first electrode elements **152** (E1-E9), and (c) the first terminal of the corresponding thermistor **154**.

(62) Notably, the plurality of thermistors **154** in this FIG. 7 embodiment are arranged in series, beginning with a first one of the thermistors (i.e., the top left in FIG. 7) and ending with a last one of the thermistors (i.e., the bottom right in FIG. 7). The second terminal of each of the thermistors except for the last thermistor is wired to the first terminal of a respective subsequent thermistor.

(63) Each transducer array **150** has a third conductor that provides an electrically conductive path between the third pin of the connector **157** and the second terminal of the last thermistor **154** (i.e., the upper terminal of the bottom right thermistor in FIG. 7). The third conductor may optionally be implemented using a plurality of segments of wire and/or a plurality of traces on a flex circuit.

(64) Because the connector **157** has an individual first pin that corresponds to each of the individual first electrode elements **152**, and because an electrically conductive path exists between each of the first pins and a respective one of the first electrode elements **152**, the system that mates with the connector **157** can selectively energize or not energize each of the first electrode elements **152** individually by either applying or not applying a signal to the respective first pin on the connector **157** during the current-outputting phase. And because the two terminals of any given one of the thermistors **154** are wired to different pins on the connector **157**, the system that mates with the connector **157** has access to both terminals of each of the thermistors **154**. As a result, the system that mates with the connector **157** can measure the resistance of any of the thermistors **154** during the temperature-reading phase.

(65) Notably, because any given first pin on the connector **157** corresponds to a respective one of the individual first electrode elements **152** (during the current-outputting phase) and also corresponds to a respective one or two of the individual thermistors **154** (during the temperature-reading phase), each of the first pins on the connector **157** serves two functions. This reduces the number of wires that must be included in each of the cables **156**, which in turn advantageously makes the cables more flexible and less cumbersome.

(66) FIG. 8 is a block diagram of a system that uses four copies of the transducer array **150** (described above in connection with FIG. 7) to apply TTFields to a subject. In FIG. 8, these four copies are labeled **150A**, **150P**, **150L**, and **150R**, where A, P, L, and R stand for anterior, posterior, left, and right, respectively. The lower portion of FIG. 8 depicts an AC voltage generator **35** and a “CAD box” **130** as separate blocks, the latter of which includes a temperature measurement block **132**, a controller **134**, and banks of switches **1L**, **2L**, **3L**, **1R**, **2R**, and **3R**. In some embodiments, the components in those two blocks **35**, **130** may be physically divided into two separate housings. But in alternative embodiments, the components in those two blocks **35**, **130** are combined into a single housing.

(67) For clarity, only the left and right channels are depicted in FIG. 8. But the remaining channels (i.e., the anterior and posterior channels) operate in the same way as the left and right channels, respectively. Additionally, each of the transducer arrays **150** in FIG. 8 is depicted with only 4 first electrode elements **152** and four thermistors **154** for purposes of clarity. But it is expected that practical systems will have a larger number (e.g., between 9 and 30) of first electrode elements and thermistors, and will also have a larger number of certain other components (e.g., switches, conductors, etc.), depending on the number of first electrode elements **152** that are actually used in each of the transducer arrays **150**.

(68) The FIG. 8 system can measure the temperatures of the thermistors **154** in the left channel **150L** during the temperature-reading phase by controlling the electronically-controlled switches in banks **2L** and **3L** (which may be implemented using bidirectional analog switches) to select each of the thermistors in turn. For example, switches **1** and **2** should be closed to select thermistor T1; switches **2** and **3** should be closed to select thermistor T2; switches **3** and **4** should be closed to select thermistor T3; and switches **4** and N should be closed to select the last thermistor (i.e., T4 in FIG. 8). After any given one of the thermistors T1-T4 within the transducer array **150L** has been

selected, the temperature measurement block **132** can determine the temperature of that thermistor by measuring the resistance of the thermistor as described above in connection with FIG. 6.

(69) Measuring the temperature of the thermistors **154** in the right channel **150R** is accomplished using the same approach described above in connection with the left channel **150L**, except that the banks of switches **2R** and **3R** are used instead of banks **2L** and **3L**. Corresponding banks of switches (not shown) are also provided for the other channels **150A**, **150P**, and a similar approach is used in those channels as well.

(70) Assume that within a given interval of time (e.g., 1 second) during the current-outputting phase, the AC voltage generator **35** is applying an AC signal between the L and R terminals. This AC signal is applied to the A terminals of the left and right transducer arrays **150L**, **150R**, which means that the output of the AC voltage generator is applied to all of the second electrode elements **153** (**A1-A4**). Based on the temperature readings obtained from the thermistors **154** (**T1-T4**) during the temperature-reading phase, the controller **134** controls the switches in banks **1L** and **1R** (and the corresponding switches in the anterior and posterior channels, not shown) to either turn on or turn off the current (which originates in the AC voltage generator **35**) to each of the corresponding first electrode elements **152** (**E1-E4**) during the next current-outputting phase as described above in connection with FIG. 6. For example, to leave the full current on for all four of the sets **152/153**, all four of the switches in bank **1L** should be closed. To reduce the current that passes through the **E1/A1** set **152/153** of electrode elements (see FIG. 7), switch **1** in bank **1L** should be opened; and to reduce the current that passes through the **E2/A2** set **152/153** of electrode elements, switch **2** in bank **1L** should be opened; etc.

(71) As in the FIG. 5-6 embodiment, controlling the current that is routed through individual first electrode elements during the current-outputting phase can be used to reduce the current that passes through a given area when that area begins to get hot; and this can advantageously prevent overheating without completely turning off the current that passes through a given area.

(72) In alternative embodiments, instead of implementing the temperature sensors using thermistors (as described above in connection with FIGS. 5-8), temperature sensing may be implemented using a plurality of regions of a pyroelectric material. Examples of suitable pyroelectric materials include PVDF homopolymers, PVDF organic derivatives e.g., P(VDF-TrFE), and PVDF-ceramic composites, where PVDF is poly(vinylidene fluoride) and TrFE is trifluoroethylene. In some embodiments, the pyroelectric material is Piezotech® RT-FC, which is a P(VDF-TrFE) copolymer. In some embodiments, the regions of pyroelectric material may be a polymer layer such as poly(vinylidene fluoride-trifluoroethylene-chlorotrifluoroethylene) and/or poly(vinylidene fluoride-trifluoroethylene-1-chlorofluoroethylene).

(73) Each of the first and second electrode elements has a front face and a rear face. Each of the regions of the pyroelectric material has a front face and a rear face, and the front face of each of the regions of the pyroelectric material is disposed in electrical and thermal contact with the rear face of the respective set of first/second electrode elements. Additional electrode regions contact the rear face of the regions of the pyroelectric material.

(74) The electrical and thermal contact between the front face of each of the regions of the pyroelectric material and the rear face of the respective set of first/second electrode elements may be achieved by placing those two faces in direct contact with each other (e.g., by depositing or spraying the first and second electrode elements onto the regions of the pyroelectric material during manufacturing). Alternatively, another layer of material that does not interrupt the electrical and thermal contact may be disposed between those two faces.

(75) Because each of the first and second electrode elements is disposed in thermal contact with a respective one of the regions of the pyroelectric material, temperature variations of any given set of first/second electrode elements will cause a corresponding change in temperature in the respective region of the pyroelectric material. This change in temperature will cause a pyroelectric voltage to appear across opposite faces of the respective region of the pyroelectric material. The instantaneous

value of this pyroelectric voltage can be measured via the first electrode elements and the additional electrode regions. And due to the thermal contact between the first and second electrode elements and the respective region of the pyroelectric material, the measured electrical signals not only represent the change in temperature of each region of the pyroelectric material—they also represent the change in temperature of each respective set of first/second electrode elements.

(76) The regions of dielectric material may be discrete sections of a flexible polymer that are separated by gaps. Alternatively, the regions of dielectric material could be regions within a single contiguous sheet of a flexible polymer material.

(77) Pyroelectric materials do not generate an output based on their absolute temperature. Instead, they generate an electrical output that is a function of a change in temperature. The first and second electrode elements are each positioned in thermal contact with a respective region of pyroelectric material with similar characteristics. So if a given set of electrode elements is running hotter than another set of electrode elements, the temperature fluctuations at the given set will be larger than the temperature fluctuations at the other set. And this will cause the region of pyroelectric material positioned in contact with the given set to generate a larger electrical output than the region of pyroelectric material positioned in contact with the other set.

(78) A controller compares the temperature fluctuations of all the electrode sets by sampling the signals that are generated by the regions of pyroelectric material that are positioned in thermal contact with those electrode sets. By analyzing how the electrical characteristics change over time, the controller can determine whether the temperature fluctuations of a given set are larger than the temperature fluctuations of another set. The controller can then use this information to normalize the temperature of all the electrode elements by switching off the first electrode element within the hotter set to reduce the current that flows through the hotter set of electrode elements.

(79) Because pyroelectric materials respond to changes in temperature (as opposed to absolute temperature), a pyroelectric material that is cycling between 37° C. and 37.2° C. will produce the same output as when that same pyroelectric material is cycling between 40° C. and a 40.2° C. In view of this, it is not sufficient to merely equalize the temperature of all the electrode elements within any given transducer array as described above. To the contrary—because the temperature of transducer arrays should be maintained below a given threshold (e.g., 40° C.), one more piece of information is needed to ensure that the transducer arrays do not overheat. And this additional piece of information is the absolute temperature of at least one set of first/second electrode elements. For if the absolute temperature of a single set of first/second electrode elements is known, and it is also known that the temperatures of all of the electrode elements have been equalized, we can then be sure that none of the electrode elements are hotter than the threshold temperature.

(80) In view of this, at least one absolute temperature sensor (e.g., a thermistor) is positioned in thermal contact with at least one set of first/second electrode elements, and the system includes a circuit so that the controller can determine the temperature of the thermistor (and thereby determine the temperature of the electrode element that is in thermal contact with the thermistor).

(81) FIG. 9 is a schematic diagram of a circuit that is suitable for implementing each of the switches in banks 1L and 1R in the FIG. 6 and FIG. 8 embodiments described above, as well as the corresponding banks for the anterior and posterior channels (not shown). The circuit includes two field effect transistors (FET) 66, 67 wired in series, which is a configuration that can pass current in either direction. One example of a suitable FET for this circuit is the BSC320N20NSE (Infineon Technologies AG, Neubiberg, Germany). (Note that the diodes depicted in FIG. 9 are inherently included within the FETs 66, 67 themselves.) The series combination of the two FETs 66, 67 will either conduct or block the flow of electricity, depending on the state of the control input that arrives from one of the digital outputs of the controller 34 described above. When the series combination is conducting, current can flow between the shared conductor and the respective first electrode element 52. On the other hand, when the series combination of FETs 66, 67 is not conducting, current will not flow between the shared conductor and the respective first electrode

element 52.

(82) In the embodiments described above in connection with FIGS. 5 and 7, all of the electrode elements 52, 53 are capacitively coupled, and the support structure 59 is configured to hold the electrode elements 52, 53 against the subject's body so that the dielectric layer of the electrode elements 52, 53 faces the subject's body and can be positioned in contact with the subject's body. But in alternative embodiments, electrode elements that are not capacitively coupled may be used. In this case, the dielectric layer of each electrode element is omitted, in which case the support structure 59 holds the electrode elements 52, 53 against the subject's body so a conductive surface of the electrode elements 52, 53 faces the subject's body and can be positioned in contact with the subject's body. Optionally, in these embodiments, a layer of hydrogel may be disposed between the conductive surface of the electrode elements 52, 53 and the subject's body when the transducer array 50 is placed against the subject's body.

(83) While the present invention has been disclosed with reference to certain embodiments, numerous modifications, alterations, and changes to the described embodiments are possible without departing from the sphere and scope of the present invention, as defined in the appended claims. Accordingly, it is intended that the present invention not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

Claims

1. An apparatus for applying an alternating electric field to a subject's body, the apparatus comprising: at least four sets of electrode elements, wherein each of the sets of electrode elements includes a respective first electrode element and a respective second electrode element disposed in thermal contact with the respective first electrode element; a connector having at least four first pins and a second pin; at least four first conductors, each of which provides an electrically conductive path between (a) a respective one of the first pins and (b) a respective one of the first electrode elements; a second conductor that provides an electrically conductive path between the second pin and all of the second electrode elements; at least four temperature sensors, each of which is disposed in thermal contact with a respective one of the sets of electrode elements; and a support configured to hold the sets of electrode elements against the subject's body.
2. The apparatus of claim 1, wherein within each of the sets of electrode elements, the area of the respective second electrode element is at least double the area of the respective first electrode element.
3. The apparatus of claim 1, wherein the apparatus has at least nine sets of electrode elements, wherein the connector has at least nine first pins, wherein the apparatus has at least nine first conductors, and wherein the apparatus has at least nine temperature sensors.
4. The apparatus of claim 1, wherein each of the temperature sensors comprises a thermistor having a first terminal and a second terminal, wherein the connector has a third pin, wherein each of the first conductors provides an electrically conductive path between (a) a respective one of the first pins, (b) a respective one of the first electrode elements, and (c) the first terminal of a respective thermistor, and wherein the apparatus further comprises a third conductor that provides an electrically conductive path between the third pin and the second terminal of at least one of the thermistors.
5. The apparatus of claim 4, wherein the second terminals of all the thermistors are wired together.
6. The apparatus of claim 4, wherein the thermistors are wired in series, beginning with a first one of the thermistors and ending with a last one of the thermistors, and wherein the second terminal of each of the thermistors except for the last thermistor is wired to the first terminal of a respective subsequent thermistor, and wherein the third conductor provides an electrically conductive path between the third pin of the connector and the second terminal of the last thermistor.

7. The apparatus of claim 1, wherein each of the temperature sensors comprises a region of a pyroelectric material.

8. The apparatus of claim 1, wherein each of the first electrode elements comprises a conductive plate with a dielectric layer disposed thereon, wherein each of the second electrode elements comprises a conductive plate with a dielectric layer disposed thereon, and wherein the support is configured to hold the first electrode elements and the second electrode elements against the subject's body so that the dielectric layers of the first electrode elements and the dielectric layers of the second electrode elements face the subject's body.
